

HDOC 17

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Batch: 16

1. Algorithm

1. Start main program.
2. Display the menu with 4 options to the user.
3. Read the user's choice.
4. Use a switch statement to call the respective function: `generateLargest()`, `generateSmallest()`, `swapFirstLast()`, or `checkPalindrome()`.
5. Function **`generateLargest()`**:
 - Read positive integer num. Initialize an array `freq[10] = {0}` and `res = 0`.
 - Extract each digit of num and increment its count in `freq`.
 - Loop ifrom 9 down to 0: while `freq[i] > 0`, append it to `res` (`res = res * 10 + i`) and decrement `freq[i]`. Display `res`.
6. Function **`generateSmallest()`**:
 - Read positive integer num. Initialize an array `freq[10] = {0}` and `res = 0`.
 - Extract each digit of num and increment its count in `freq`.
 - Find the first non-zero digit (from 1 to 9), append it to `res`, and decrement its frequency.
 - Loop ifrom 0 to 9: while `freq[i] > 0`, append it to `res` and decrement `freq[i]`. Display `res`.
7. Function **`swapFirstLast()`**:
 - Read positive integer num. If `num < 10`, the number remains unchanged.
 - Find the number of digits. Calculate a multiplier `mult = 10^(digits-1)`.

- Extract first = num / mult and last = num % 10. Extract the middle part: middle = (num % mult) / 10.
- Construct swapped number: newNum = last * mult + middle * 10 + first. Display newNum.

8. Function **checkPalindrome()**:

- Read positive integer num. Initialize temp = num and rev = 0.
- Loop while temp > 0: rev = rev * 10 + temp % 10, then temp = temp / 10.
- If rev == num, display "Palindrome". Else, display "Not a Palindrome".

9. Stop

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num)	Expected Output
TC-01	Largest possible number	Choice: 1 Num: 2753	Largest Possible: 7532
TC-02	Smallest possible number (handles zeros)	Choice: 2 Num: 27503	Smallest Possible: 20357
TC-03	Swap first and last digits	Choice: 3 Num: 12345	After Swap: 52341
TC-04	Palindrome check (True)	Choice: 4 Num: 12321	Output: Palindrome
TC-05	Palindrome check (False)	Choice: 4 Num: 1234	Output: Not a Palindrome
TC-06	Invalid choice	Choice: 5	Invalid choice.

3. C Program Source Code

```
#include <stdio.h>

// Function Prototypes (void return type, no arguments) void generateLargest(void);
void generateSmallest(void); void
swapFirstLast(void); void
checkPalindrome(void);

int main() {
    int choice;
    printf("--- HDOC 17 Menu ---\n");
    printf("1. Generate the largest possible number\n"); printf("2. Generate the
smallest possible number\n"); printf("3. Swap first and last digits\n");
    printf("4. Check whether it is a palindrome\n"); printf("Enter your choice (1-4):
");
    scanf("%d", &choice);

    switch (choice) {
        case 1: generateLargest(); break; case 2:
generateSmallest(); break; case 3: swapFirstLast();
break; case 4: checkPalindrome(); break;
        default: printf("Invalid choice.\n"); break;
    }
    return 0;
}

// Function 1: Generate Largest void
generateLargest(void) {
    long long num, temp, res = 0; int freq[10] =
{0}, i;

    printf("Enter a positive integer: "); scanf("%lld", &num);
    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    temp = num;
    while (temp > 0) { freq[temp % 10]++; temp /= 10; }

    for (i = 9; i >= 0; i--) { while (freq[i] > 0) {
        res = res * 10 + i; freq[i]--;
    }
    }
    printf("Number: %lld, Largest Possible: %lld\n", num, res);
}
```

```

// Function 2: Generate Smallest void
generateSmallest(void) {
    long long num, temp, res = 0; int freq[10] =
    {0}, i;

    printf("Enter a positive integer: "); scanf("%lld", &num);
    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    temp = num;
    while (temp > 0) { freq[temp % 10]++; temp /= 10; }

    // Find first non-zero digit to place at the highest magnitude for (i = 1; i <= 9; i++) {
        if (freq[i] > 0) { res = i;
            freq[i]--; break;
        }
    }

    // Append the remaining digits (including zeros) for (i = 0; i <= 9; i++) {
        while (freq[i] > 0) { res = res * 10
            + i; freq[i]--;
        }
    }
    printf("Number: %lld, Smallest Possible: %lld\n", num, res);
}

```

```

// Function 3: Swap First and Last void
swapFirstLast(void) {
    long long num, temp, newNum, mult = 1, middle; int first, last, digits =
    0, i;

    printf("Enter a positive integer: "); scanf("%lld", &num);
    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    if (num < 10) {
        printf("Number: %lld, After Swap: %lld\n", num, num); return;
    }

    temp = num;
    while (temp > 0) { digits++; temp /= 10; } for (i = 1; i < digits;

    i++) mult *= 10; first = num / mult;

```

```

    last = num % 10;
    middle = (num % mult) / 10;

    newNum = last * mult + middle * 10 + first; printf("Number: %lld, After
    Swap: %lld\n", num, newNum);
}

// Function 4: Check Palindrome void
checkPalindrome(void) {
    long long num, temp, rev = 0;

    printf("Enter a positive integer: "); scanf("%lld", &num);
    if (num <= 0) { printf("Error: Positive integer required.\n"); return; }

    temp = num;
    while (temp > 0) {
        rev = rev * 10 + temp % 10; temp /= 10;
    }

    if (num == rev) printf("Number: %lld, Output: Palindrome\n", num); else
    printf("Number: %lld, Output: Not a Palindrome\n", num);
}

```